

Distribution XML reference for Warteschlangensimulator

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This reference refers to version 6.0.0 of Warteschlangensimulator.
Download address: <https://a-herzog.github.io/Warteschlangensimulator/>.

When storing distribution settings to xml files the following xml tags will be used:

English version:

```
<ModelElementDistribution>distribution name (parameters)</ModelElementDistribution>
```

German version:

```
<ModellElementVerteilung>distribution name (parameters)</ModellElementVerteilung>
```

The English or German version will be used when storing xml data in Warteschlangensimulator. When reading xml files Warteschlangensimulator will always understand both versions.

In the following sections the possible values for "distribution name" and the corresponding "parameters" will be listed. Distribution parameters "mean" and "sd" correspond directly to the stochastic characteristics mean and standard deviation.

Empirical data

Empirical data (point1;point2;point3;...)

Empirische Daten (point1;point2;point3;...)

The data points are interpreted as pdf values at equidistant points on the predefined range of the distribution.

One point distribution

One point distribution (point)

Ein-Punkt-Verteilung (point)

Conversion between distribution parameters and stochastic characteristics

$$mean = point$$

$$sd = 0$$

Uniform distribution

Uniform distribution (lower;upper)

Gleichverteilung (lower;upper)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
 mean &= (lower + upper)/2 \\
 sd &= (upper - lower)/\sqrt{12} \\
 lower &= mean - sd \cdot \sqrt{12}/2 \\
 upper &= mean + sd \cdot \sqrt{12}/2
 \end{aligned}$$

Exponential distribution

Exponentialverteilung (mean)
 Exponential distribution (mean)

Conversion between distribution parameters and stochastic characteristics

$$sd = mean$$

Normal distribution

Normal distribution (mean;sd)
 Normalverteilung (mean;sd)

Lognormal distribution

Lognormal distribution (mean;sd)
 Lognormalverteilung (mean;sd)

Erlang distribution

Erlang distribution (shape;scale)
 Erlang-Verteilung (shape;scale)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
 mean &= shape \cdot scale \\
 sd &= \sqrt{shape} \cdot scale \\
 scale &= sd^2 / mean \\
 shape &= mean^2 / sd^2
 \end{aligned}$$

Gamma distribution

Gamma distribution (shape;scale)
 Gamma-Verteilung (shape;scale)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= shape \cdot scale \\
sd &= \sqrt{shape} \cdot scale \\
scale &= sd^2 / mean \\
shape &= mean^2 / sd^2
\end{aligned}$$

Beta distribution

Beta distribution (alpha;beta;lower;upper)

Beta-Verteilung (alpha;beta;lower;upper)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= alpha / (alpha + beta) \cdot (upper - lower) + lower \\
sd &= (upper - lower)^2 \cdot alpha \cdot beta / (alpha + beta)^2 / (1 + alpha + beta)
\end{aligned}$$

Cauchy distribution

Cauchy-Verteilung (median;scale)

Cauchy distribution (median;scale)

Half Cauchy distribution

Halbe Cauchy-Verteilung (mu;sigma)

Half Cauchy distribution (mu;sigma)

Weibull distribution

Weibull distribution (scaleInvers;shape)

Weibull-Verteilung (scaleInvers;shape)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= scale \cdot \Gamma(1 + (1/shape)) \\
sd &= scale \cdot \sqrt{\Gamma(1 + 2/shape) - (\Gamma(1 + 1/shape))^2}
\end{aligned}$$

Chi distribution

Chi distribution (degreesOfFreedom)

Chi-Verteilung (degreesOfFreedom)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= \sqrt{2} \cdot \Gamma((degreesOfFreedom + 1)/2) / \Gamma(degreesOfFreedom/2) \\
sd &= \left[(2 \cdot \Gamma(degreesOfFreedom/2) * \Gamma(1 + degreesOfFreedom/2) - \right. \\
&\quad \left. (\Gamma((degreesOfFreedom + 1)/2))^2) / \Gamma(degreesOfFreedom/2) \right]^{\frac{1}{2}}
\end{aligned}$$

Chi² distribution

Chi² distribution (degreesOfFreedom)

Chi²-Verteilung (degreesOfFreedom)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \text{degreesOfFreedom} \\ \text{sd} &= \sqrt{2 \cdot \text{degreesOfFreedom}} \end{aligned}$$

F distribution

F distribution (NumeratorDegreesOfFreedom;DenominatorDegreesOfFreedom)

F-Verteilung (NumeratorDegreesOfFreedom;DenominatorDegreesOfFreedom)

Conversion between distribution parameters and stochastic characteristics

(let $m := \text{NumeratorDegreesOfFreedom}$ and $n := \text{DenominatorDegreesOfFreedom}$)

$$\begin{aligned} \text{mean} &= \frac{n}{n-2} \\ \text{sd} &= \sqrt{2 \cdot n^2 \cdot \frac{m+n-2}{m \cdot (n-2) \cdot (n-2) \cdot (n-4)}} \\ m &= \text{round} \left(2 \cdot (2 \cdot \tilde{m})^2 \cdot \frac{2 \cdot \tilde{m} - 2}{\text{sd}^2 \cdot (2 \cdot \tilde{m} - 2)^2 \cdot (2 \cdot \tilde{m} - 4) - 2 \cdot (2 \cdot \tilde{m})^2} \right) \text{ with } \tilde{m} := \text{mean}/(\text{mean} - 1) \\ n &= \text{round}(2 \cdot \text{mean}/(\text{mean} - 1)) \end{aligned}$$

Johnson SU distribution

Johnson SU distribution (gamma;xi;delta;lambda)

Johnson-SU-Verteilung (gamma;xi;delta;lambda)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \text{xi} - \text{lambda} \cdot \exp(1/(2 \cdot \text{delta}^2)) \cdot \sinh(\text{gamma}/\text{delta}) \\ \text{sd} &= \sqrt{\text{lambda}^2/2 \cdot (\exp(1/\text{delta}^2) - 1) \cdot (\exp(1/\text{delta}^2) * \cosh(2 * \text{gamma}/\text{delta}) + 1)} \end{aligned}$$

Triangular distribution

Triangular distribution (lowerBound;mostLikelyX;upperBound)

Dreiecksverteilung (lowerBound;mostLikelyX;upperBound)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= (lowerBound + mostLikelyX + upperBound)/3 \\
sd &= \left[(lowerBound^2 + upperBound^2 + mostLikelyX^2 - lowerBound \cdot upperBound - \right. \\
&\quad \left. lowerBound \cdot mostLikelyX - upperBound \cdot mostLikelyX)/18 \right]^{\frac{1}{2}} \\
lowerBound &= mean - sd \cdot \sqrt{6} \\
mostLikelyX &= mean \\
upperBound &= mean + sd \cdot \sqrt{6}
\end{aligned}$$

Trapezoid distribution

Trapezoid distribution (a;b;c;d)

Trapezverteilung (a;b;c;d)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
h &= \frac{2}{c + d - a - b} \\
mean &= h \cdot 16 \left(\frac{d^3 - c^3}{d - c} - \frac{b^3 - a^3}{b - a} \right) \\
sd &= \sqrt{h \cdot 112 \left(\frac{d^4 - c^4}{d - c} - \frac{b^4 - a^4}{b - a} \right) - mean^2}
\end{aligned}$$

Pert distribution

Pert distribution (lowerBound;mostLikelyX;upperBound)

Pert-Verteilung (lowerBound;mostLikelyX;upperBound)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= (lowerBound + 4 \cdot mostLikelyX + upperBound)/6 \\
sd &= \left[((lowerBound + 4 \cdot mostLikelyX + upperBound)/6 - lowerBound) \cdot \right. \\
&\quad \left. (upperBound - (lowerBound + 4 \cdot mostLikelyX + upperBound)/6)/7 \right]^{\frac{1}{2}} \\
lowerBound &= mean - sd \cdot \sqrt{7} \\
mostLikelyX &= mean \\
upperBound &= mean + sd \cdot \sqrt{7}
\end{aligned}$$

Laplace distribution

Laplace distribution (mean;b)

Laplace-Verteilung (mean;b)

Conversion between distribution parameters and stochastic characteristics

$$sd = b \cdot \sqrt{2}$$

Pareto distribution

Pareto distribution (xmin;alpha)

Pareto-Verteilung (xmin;alpha)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} mean &= alpha \cdot xmin / (alpha - 1) \\ sd &= xmin^2 \cdot alpha / (alpha - 1)^2 / (alpha - 2) \\ alpha &= mean / (mean - xmin) \end{aligned}$$

Logistic distribution

Logistic distribution (mean;s)

Logistische Verteilung (mean;s)

Conversion between distribution parameters and stochastic characteristics

$$sd = s \cdot \pi / \sqrt{3}$$

Inverse gaussian distribution

Inverse gaussian distribution (lambda;mu)

Inverse Gauß-Verteilung (lambda;mu)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} mean &= mu \\ sd &= mu \cdot \sqrt{mu / lambda} \\ lambda &= mean^3 / sd^2 \end{aligned}$$

Rayleigh distribution

Rayleigh distribution (mean)

Rayleigh-Verteilung (mean)

Conversion between distribution parameters and stochastic characteristics

$$sd = \sqrt{(4 - \pi)/2} \cdot \sqrt{2/\pi} \cdot mean$$

Log-logistic distribution

Log-logistic distribution (alpha;beta)

Log-Logistische Verteilung (alpha;beta)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} mean &= alpha \cdot \pi / beta / \sin(\pi / beta) \\ sd &= alpha \cdot \sqrt{2 \cdot \pi / beta / \sin(2 \cdot \pi / beta) - \pi^2 / beta^2} / \sin(\pi / beta) \end{aligned}$$

Power distribution

Power distribution (a;b;c)

Potenzverteilung (a;b;c)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= a + (b - a) \cdot c / (c + 1) \\ \text{sd} &= (b - a) / (c + 1) \cdot \sqrt{c / (c + 2)} \end{aligned}$$

Gumbel distribution

Gumbel distribution (mean;sd)

Gumbel-Verteilung (mean;sd)

Fatigue life distribution

Fatigue life distribution (mu;beta;gamma)

Fatigue-Life-Verteilung (mu;beta;gamma)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \mu + \beta \cdot (1 + \gamma \cdot \gamma / 2) \\ \text{sd} &= \beta \cdot \gamma \cdot \sqrt{1 + 5 \cdot \gamma^2 / 4} \end{aligned}$$

Frechet distribution

Frechet distribution (delta;beta;alpha)

Frechet-Verteilung (delta;beta;alpha)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \delta + \beta \cdot \gamma (1 - 1/\alpha) \\ \text{sd} &= \beta \cdot \sqrt{\gamma (1 - 2/\alpha) - \gamma (1 - 1/\alpha)^2} \end{aligned}$$

Hyperbolic secant distribution

Hyperbolic secant distribution (mean;sd)

Hyperbolische Sekanten-Verteilung (mean;sd)

Left sawtooth distribution

Left sawtooth distribution (a;b)

Linke Sägezahnverteilung (a;b) Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= (2 \cdot a + b)/2 \\ \text{sd} &= (b - a)^2/18 \end{aligned}$$

Right sawtooth distribution

Right sawtooth distribution (a;b)

Rechte Sägezahnverteilung (a;b) Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= (a + 2 \cdot b)/2 \\ \text{sd} &= (b - a)^2/18 \end{aligned}$$

Levy distribution

Levy distribution (mu;c)

Levy-Verteilung (mu;c)

Maxwell Boltzmann distribution

Maxwell Boltzmann distribution (a)

Maxwell-Boltzmann-Verteilung (a)

$$\begin{aligned} \text{mean} &= 2 \cdot a \sqrt{\frac{2}{\pi}} \\ \text{sd} &= \sqrt{\frac{a^2(3\pi - 8)}{\pi}} \\ \text{modus} &= a\sqrt{2} \\ a &= \frac{\text{mean}}{2} \cdot \sqrt{\frac{\pi}{2}} \end{aligned}$$

Student t-distribution

Student t-distribution (mu;nu)

Studentsche t-Verteilung (mu;nu) Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \mu \\ \text{sd} &= \sqrt{\frac{\nu}{\nu - 2}} \quad \text{for } \nu > 2 \end{aligned}$$

Half normal distribution

Halbe Normalverteilung (s;mu)

Half normal distribution (s;mu)

Conversion between distribution parameters and stochastic characteristics

$$\theta = \frac{1}{\mu}$$

$$mean = \frac{1}{\theta} + s$$

$$sd = \frac{\pi - 2}{2\theta^2}$$

U-quadratic distribution

U-quadratische Verteilung (a;b)

U-quadratic distribution (a;b)

Conversion between distribution parameters and stochastic characteristics

$$mean = \frac{a + b}{2}$$

$$sd = \frac{3}{20} \cdot (b - a)^2$$

$$a = mean - \frac{\sqrt{sd^2 \cdot \frac{20}{3}}}{2}$$

$$b = mean + \frac{\sqrt{sd^2 \cdot \frac{20}{3}}}{2}$$

Reciprocal distribution

Reziproke Verteilung (a;b)

Reciprocal distribution (a;b)

Conversion between distribution parameters and stochastic characteristics

$$mean = \frac{b - a}{\log\left(\frac{b}{a}\right)}$$

$$sd = \sqrt{\frac{b^2 - a^2}{\log\left(\frac{b}{a}\right)} - \left(\frac{b - a}{\log\left(\frac{b}{a}\right)}\right)^2}$$

Kumaraswamy distribution

Kumaraswamy-Verteilung (a;b;c;d)

Kumaraswamy distribution (a;b;c;d)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean_{\text{unscaled}} &= \frac{b\Gamma\left(1 + \frac{1}{a}\right)\Gamma(b)}{\Gamma\left(1 + \frac{1}{a} + b\right)} \\
sd_{\text{unscaled}} &= \sqrt{\frac{b\Gamma\left(1 + \frac{2}{a}\right)\Gamma(b)}{\Gamma\left(1 + \frac{2}{a} + b\right)} - mean_{\text{unscaled}}^2} \\
mean &= mean_{\text{unscaled}} \cdot (d - c) + c \\
sd &= sd_{\text{unscaled}} \cdot (d - c) + c
\end{aligned}$$

Irwin-Hall distribution

Irwin-Hall-Verteilung (n)
 Irwin-Hall distribution (n)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= \frac{n}{2} \\
sd &= \sqrt{\frac{n}{12}}
\end{aligned}$$

Sine distribution

Sinus-Verteilung (a;b)
 sine distribution (a;b)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= \frac{a + b}{2} \\
sd &= \sqrt{\frac{1}{4} - \frac{2}{\pi^2}} \cdot (b - a)^2
\end{aligned}$$

Cosine distribution

Cosinus-Verteilung (a;b)
 cosine distribution (a;b)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
mean &= \frac{a + b}{2} \\
sd &= (b - a) \sqrt{\frac{\pi^2 - 6}{12\pi^2}}
\end{aligned}$$

Arcsine distribution

Arcus Sinus-Verteilung (a;b)
 arcsine distribution (a;b)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \frac{a+b}{2} \\ \text{sd} &= \frac{1}{8} \cdot (b-a)^2 \end{aligned}$$

Wigner half-circle distribution

Wigner Halbkreis-Verteilung (m;R)
Wigner half-circle distribution (m;R)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= m \\ \text{sd} &= \frac{R}{2} \end{aligned}$$

Log-Cauchy distribution

Log-Cauchy-Verteilung (mu;sigma)
Log-Cauchy distribution (mu;sigma)

Conversion between distribution parameters and stochastic characteristics

$$\text{median} = \exp(mu)$$

Log-Gamma distribution

Log-Gamma-Verteilung (a;b)
Log-Gamma distribution (a;b)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \left(1 - \frac{1}{b}\right)^{-a} \\ \text{st} &= \sqrt{\left(1 - \frac{2}{b}\right)^{-a} - \left(1 - \frac{1}{b}\right)^{-2a}} \end{aligned}$$

Log-Laplace distribution

Log-Laplace-Verteilung (c;s)
Log-Laplace distribution (c;s)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= s + \frac{c^2}{(c-1)(c+1)} \\ \text{st} &= \sqrt{\frac{c^2}{(c-2)(c+2)} - \frac{c^4}{(c-1)^2(c+1)^2}} \end{aligned}$$

Geometric distribution

Geometric distribution (p) **Geometrische Verteilung (p)** Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \frac{1-p}{p} \\ \text{sd} &= \sqrt{\frac{1-p}{p^2}} \\ p &= \frac{1}{1+\text{mean}} \end{aligned}$$

Hypergeometric distribution

Hypergeometric distribution (N;K;n) **Hypergeometrische Verteilung (N;K;n)** Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \frac{n \cdot K}{N} \\ \text{sd} &= \sqrt{\frac{n \cdot K}{N} \cdot \left(1 - \frac{K}{N}\right) \cdot \frac{N-n}{N-1}} \end{aligned}$$

Binomial distribution

Binomial distribution (n;p) **Binomialverteilung (n;p)** Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= n \cdot p \\ \text{sd} &= \sqrt{n \cdot p \cdot (1-p)} \\ n &= \frac{\text{mean}}{p} \\ p &= 1 - \frac{\text{sd}^2}{\text{mean}} \end{aligned}$$

Poisson distribution

Poisson distribution (lambda) **Poisson-Verteilung (lambda)**

$$\begin{aligned} \text{mean} &= \lambda \\ \text{sd} &= \lambda \end{aligned}$$

Negative binomial distribution

Negative binomial distribution (n;r) **Negative Binomialverteilung (n;r)**

$$\begin{aligned}
 mean &= r(1-p)/p \\
 sd &= \sqrt{\frac{r(1-p)}{p^2}} \\
 p &= \frac{mean}{sd^2} \\
 r &= \frac{mean \cdot p}{1-p}
 \end{aligned}$$

Negative Hypergeometric distribution

Negative hypergeometric distribution (N;K;n) Negative hypergeometrische Verteilung (N;K;n)
Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned}
 mean &= n \cdot \frac{N+1}{R+1} \\
 sd &= \sqrt{n \cdot \frac{(R+1-n)(N-R)(N+1)}{(R+1)^2(R+2)}}
 \end{aligned}$$

Zeta distribution

Zeta distribution (s) Zeta-Verteilung (s)

$$\begin{aligned}
 mean &= \frac{\zeta(s-1)}{\zeta(s)} \\
 sd &= \sqrt{\frac{\zeta(s)\zeta(s-2) - \zeta(s-1)^2}{\zeta(s)^2}}
 \end{aligned}$$

Discrete uniform distribution

Discrete uniform distribution (a;b) Diskrete Gleichverteilung (n;r)

$$\begin{aligned}
 mean &= \frac{a+b}{2} \\
 sd &= \frac{(b-a+1)^2 - 1}{12} \\
 b &= \frac{\sqrt{(12 \cdot sd^2 + 1)}}{2} + mean - \frac{1}{2} \\
 a &= 2 \cdot mean - b
 \end{aligned}$$

Logarithmic distribution

Logarithmic distribution (p) Logarithmische Verteilung (p) Conversion between distribution parameters and stochastic characteristics

$$mean = -\frac{p}{(1-p) \log(1-p)}$$

Borel distribution

Borel distribution (mu) **Borel-Verteilung (mu)** Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \frac{1}{1 - \mu} \\ \text{variance} &= \frac{\mu}{(1 - \mu)^3} \end{aligned}$$

Planck distribution

Planck distribution (lambda) **Planck-Verteilung (lambda)**

$$\begin{aligned} \text{mean} &= \frac{1}{\exp(\lambda) - 1} \\ \text{sd} &= \frac{\sqrt{\exp(-\lambda)}}{1 - \exp(-\lambda)} \end{aligned}$$

Inverse Gamma distribution

Inverse gamma distribution (shape;scale)

Inverse Gamma-Verteilung (shape;scale)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= \frac{\text{scale}}{\text{shape} - 1} \\ \text{sd} &= \frac{\text{scale}}{(\text{shape} - 1)\sqrt{\text{shape} - 2}} \\ \text{shape} &= \frac{\text{mean}^2}{\text{sd}^2} + 2 \\ \text{scale} &= \frac{\text{mean}^3}{\text{sd}^2} + \text{mean} \end{aligned}$$

Generalized Rademacher distribution

Generalized Rademacher distribution (a;b;pa)

Verallgemeinerte Rademacher-Verteilung (a;b;pa)

Conversion between distribution parameters and stochastic characteristics

$$\begin{aligned} \text{mean} &= a \cdot P(a) + b \cdot (1 - P(a)) \\ \text{sd} &= \sqrt{a^2 \cdot P(a) + b^2 \cdot (1 - P(a)) - \text{mean}^2} \end{aligned}$$

Continuous Bernoulli distribution

Continuous Bernoulli distribution (a;b;lambda)

Kontinuierliche Bernoulli-Verteilung (a;b;lambda)

Conversion between distribution parameters and stochastic characteristics

$$mean = \frac{\lambda}{2\lambda - 1} + \frac{1}{2 \tanh^{-1}(1 - 2\lambda)} \text{ if } \lambda \neq \frac{1}{2} \text{ for } \lambda = \frac{1}{2} : 2$$

$$sd = \sqrt{-\frac{(1 - \lambda)\lambda}{(1 - 2\lambda)^2} + \frac{1}{(2 \tanh^{-1}(1 - 2\lambda))^2}} \text{ if } \lambda \neq \frac{1}{2} \text{ for } \lambda = \frac{1}{2} : \frac{1}{\sqrt{2}}$$

Boltzmann distribution

Boltzmann distribution (lambda;N) Boltzmann-Verteilung (lambda;N)

$$mean = \frac{\exp(-\lambda)}{1 - \exp(-\lambda)} - N \cdot \frac{\exp(-\lambda N)}{1 - \exp(-\lambda N)}$$

$$sd = \sqrt{\frac{\exp(-\lambda)}{(\exp(-\lambda))^2} - N^2 \cdot \frac{\exp(-\lambda N)}{(1 - \exp(-\lambda N))^2}}$$